SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	S. Mahammed Nayeem	**Reg. No.:**	192425294
Section / Batch:	AI&ML/2024	**Date:**	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:		**100 Marks**

SECTION A – CONCEPT MAPPING

Q1.Explain how operation count depends on input size and leads to time complexity formulation. Extend the map by including Growth Function as an intermediate node linking operation count and asymptotic notation. Justify how asymptotic notation simplifies comparison of algorithms for large inputs. Interpret how this mapping helps evaluate efficiency across different problem sizes.

Q2.Explain how time and space complexities are related. Extend the map by including performance and memory usage nodes. Justify why trade-offs are necessary. Interpret how this mapping supports balanced algorithm design.

Code of Conduct

I S.Mohammed Nayeem/192425994 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______	Signature: ______	Signature: ______
Date: ______	Date: ______	Date: ______

CO1

1.

Operation count and Time Complexity:-

Introduction:

The efficiency of an algorithm depends on the number of operations it performs for a given input size (n). As input size increases, the operation count also increases, affecting the algorithm's execution time.

Concept Map:-

Algorithm
(step by step procedure

↓

Input size(n)
(size of the problem)

↓

Operation Count
(Number of basic operations)

↓

Growth Function f(n)
(Describe how operation count)

↓

Asymtotic Notation (O, Ω, θ)
(Represents growth rate)

↓

Time Complexity ($T(n)$)
(Measure running time for large n)

Efficiency Evaluation
(Compare algorithms for different n)

* Linear search $\rightarrow$ n comparisions $\rightarrow$ $O(n)$
* Binary search $\rightarrow$ log n comparisions $\rightarrow$ $O\log(n)$
* Bubble sort $\rightarrow$ n^2 comparisions $\rightarrow$ $O(n^2)$

<u>Examples:-</u>

1) * $f(n) = n$
   * $f(n) = n\log n$
   * $f(n) = n^2$
   * $f(n) = 2^n$

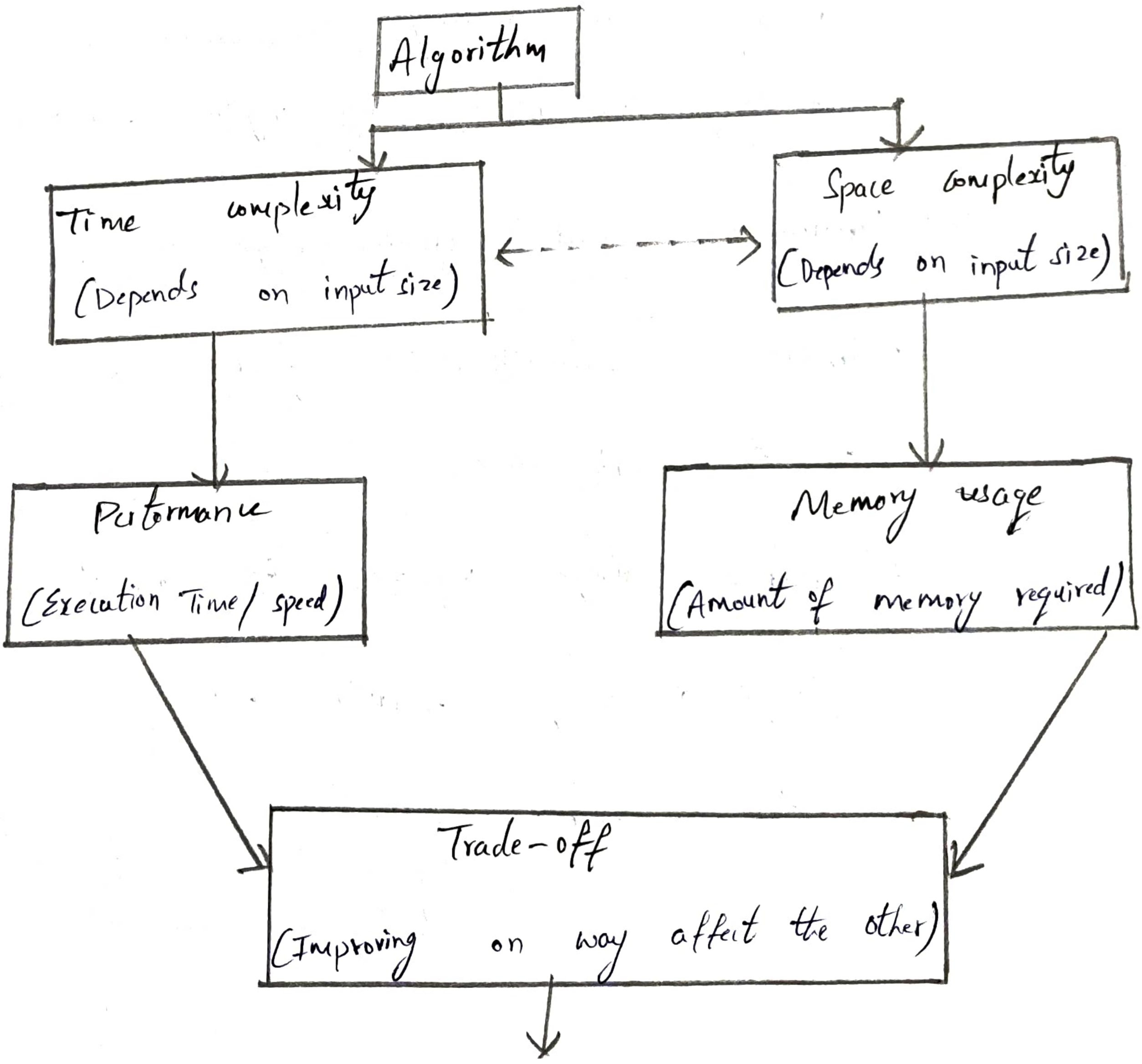
2) * $3n + 5 \rightarrow O(n)$
   * $2n^2 + 7n + 4 \rightarrow O(n^2)$
   * $5\log n + 8 \rightarrow O(\log n)$

Performance and memory usage as Intermediate Nodes:-

* Performance represents how quickly an algorithm completes its task and is directly influenced by time complexity.

* Memory usage represents the amount of storage required during execution and is directly related to space complexity.

Concept Map:-



Balanced Algorithm Design
(optimal balance between time and space for given constraints)

1. Relation between Time and space Complexity:-

Time complexity measures the amount of execution time an algorithm as the input size increases.

For example,

* Using extra memory can reduce execution time.
* Using less money may require additional computational.

2. Performance and memory usage as Intermediate Nodes:-

* Performance represents how quickly an algorithm completes it task.

* Memory usage represents the amount of storage required during execution and is directly related to space complexity.